



Association-Based Spectral Clustering for Mixed Data

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Distance-based learning with mixed data

Line of work: increasing **awareness** in distance construction

Scale/type-aware

Additivity, commensurability, and bias in mixed data

Association-aware

Redundancy, correlations, and categorical associations

Interaction-aware

Continuous–categorical relationships and local neighbourhood structure

Response-aware

Supervised neighbourhoods and prediction-oriented distances



Building distance-based pipelines: **manydist**¹

Distance construction, diagnostics, and learning workflows

Today's focus

Line of work: increasing **awareness** in distance construction

Scale/type-aware

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Supervised neighbourhoods and prediction-oriented distances

↪ **Carlo's talk**



Building distance-based pipelines: **manydist**¹

Distance construction, diagnostics, and learning workflows

↪ **Angelos' talk**

setup

Learning from distances

Many learning methods can operate directly on a dissimilarity matrix.

Dimension reduction

Multidimensional scaling (MDS)¹

Clustering methods

Hierarchical clustering (HC),
PAM², and **spectral clustering**³

Nearest-neighbour prediction

Nearest-neighbour classification
and nearest-neighbour
averaging for regression

choosing a different **distance** can lead to a different downstream analysis result

Mixed-data setup

a mixed data set

- I observations described by Q variables, Q_n numerical and Q_c categorical
- the $I \times Q$ data matrix $\mathbf{X} = [\mathbf{X}_n, \mathbf{X}_c]$ is column-wise partitioned

A formulation for mixed distance between observations i and ℓ :

$$\begin{aligned} d(\mathbf{x}_i, \mathbf{x}_\ell) &= \sum_{j_n=1}^{Q_n} d_{j_n}(\mathbf{x}_i^n, \mathbf{x}_\ell^n) + \sum_{j_c=1}^{Q_c} d_{j_c}(\mathbf{x}_i^c, \mathbf{x}_\ell^c) = \\ &= \sum_{j_n=1}^{Q_n} w_{j_n} \delta_{j_n}^n(\mathbf{x}_i^n, \mathbf{x}_\ell^n) + \sum_{j_c=1}^{Q_c} w_{j_c} \delta_{j_c}^c(\mathbf{x}_i^c, \mathbf{x}_\ell^c) \end{aligned}$$

a by-variable weighted sum of dissimilarities

Independence-based distances

Independence-based pairwise distance

No inter-variable relations are considered.

- in the continuous case: **Euclidean** or **Manhattan** distances
- in the categorical case: **Hamming** / matching distance, among many others
- in the mixed-data case: **Gower** dissimilarity index

Beyond commensurability

commensurability makes variable contributions comparable across **scales** and data **types**¹.

- If variables are correlated or associated, the same information may contribute repeatedly to the distance: **redundancy**

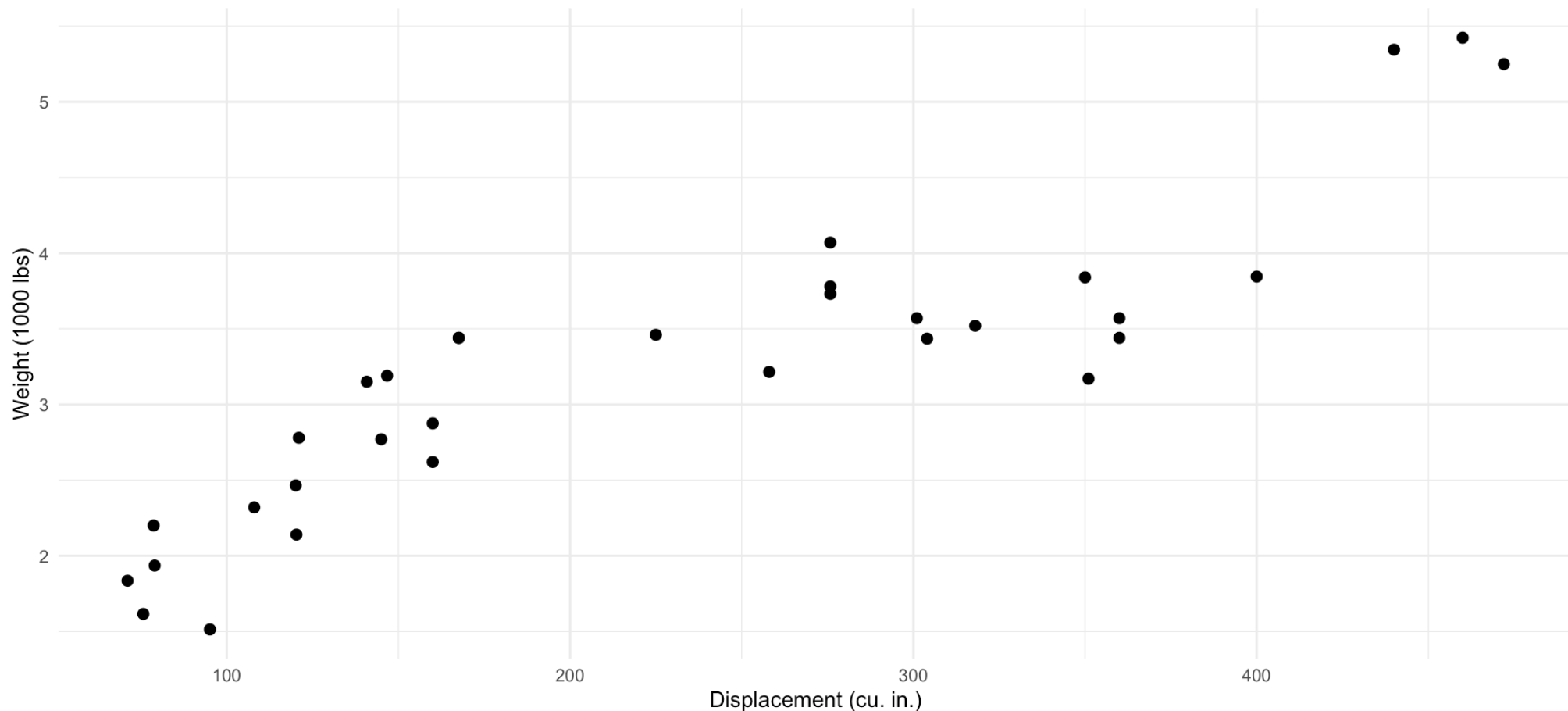
Association-aware distances

by variable differences: independence-based

- When variables are correlated or associated, shared information is effectively counted multiple times
- inflated dissimilarities may cause potential distortions in downstream unsupervised learning tasks.

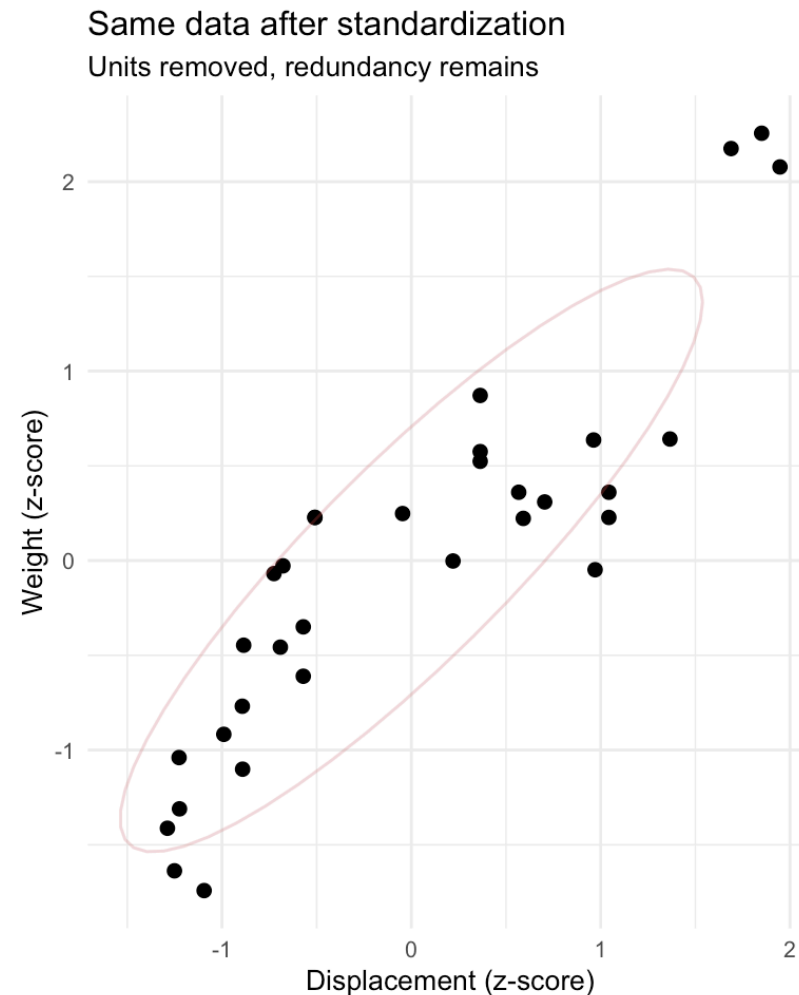
Redundant continuous variables: displacement and weight

Larger cars tend to have both higher displacement and weight



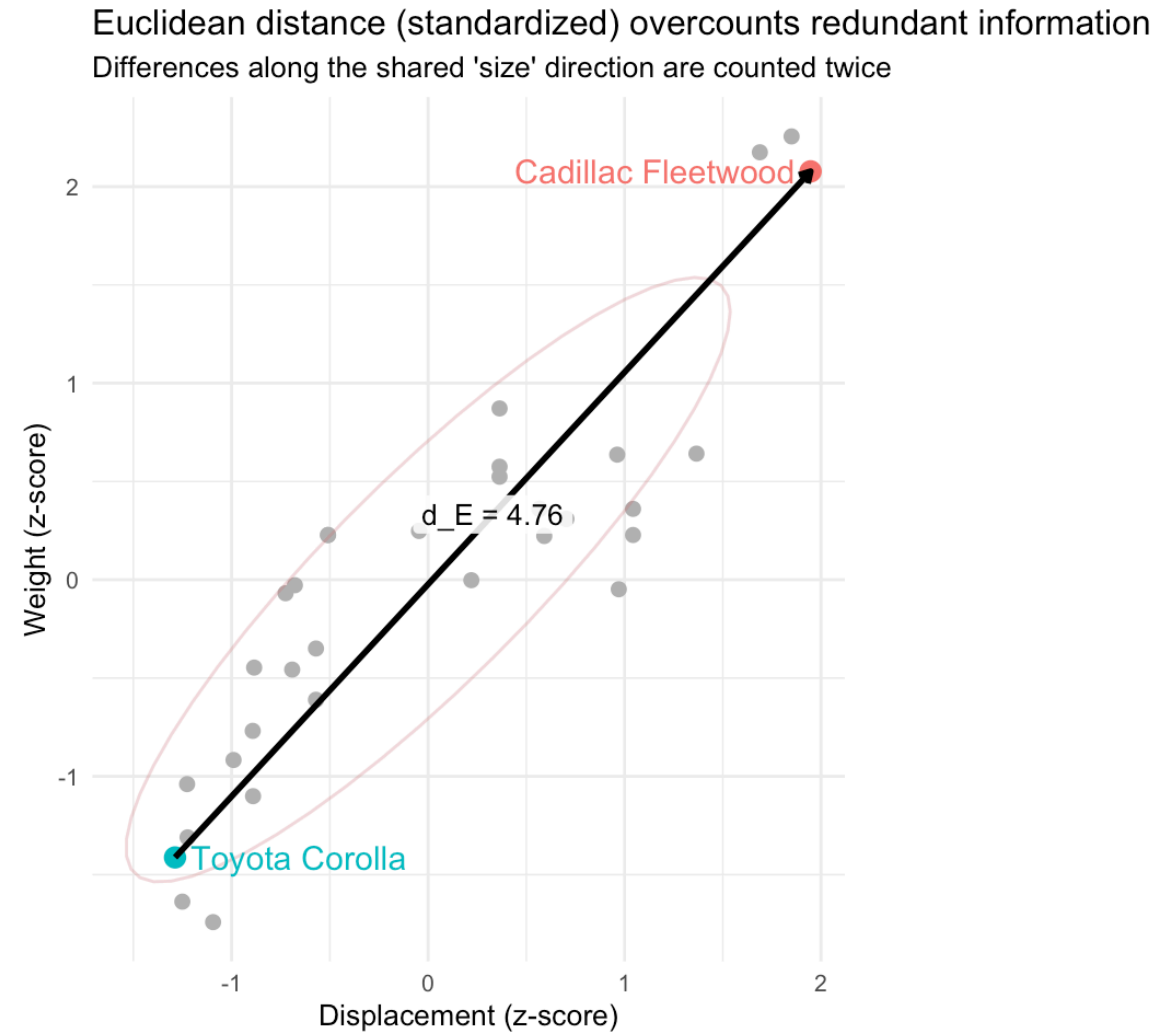
by variable differences: **independence-based**

- When variables are correlated or associated, shared information is effectively counted multiple times
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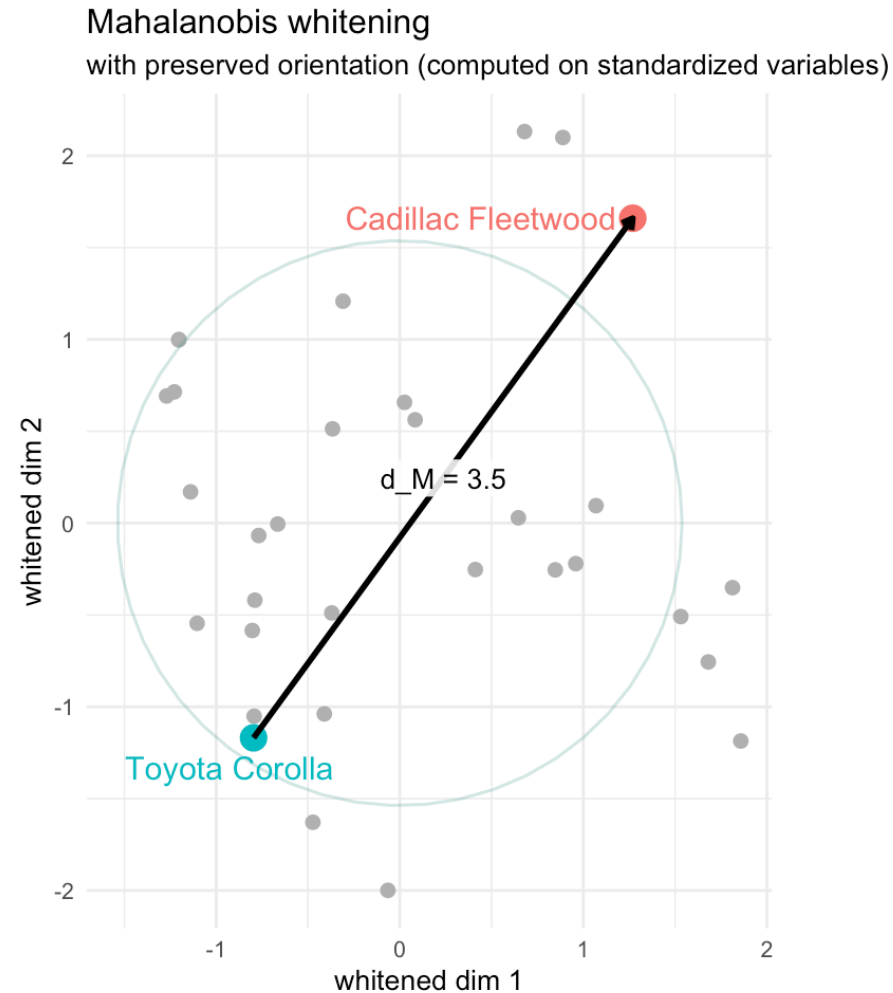
by variable differences: independence-based

The Euclidean distance $\longrightarrow$ shared information is over-counted



accounting for inter-variable relations: **association-based**

The **Mahalanobis distance** → shared information is not over-counted



this is an **association-based** distance for continuous data

association-based distance

Association-based for continuous: Mahalanobis distance

Let $\mathbf{X}_n$ be $I \times Q_n$ a data matrix of I observations described by Q_n continuous variables, and let $\mathbf{S}$ the sample covariance matrix, the Mahalanobis distance matrix is

$$\mathbf{D}_{mah} = [\text{diag}(\mathbf{G}) \mathbf{1}^\top + \mathbf{1} \text{diag}(\mathbf{G})^\top - 2\mathbf{G}]^{\odot 1/2}$$

where

- $[\cdot]^{\odot 1/2}$ denotes the element-wise square root and $\mathbf{1}$ an I -dimensional vector of 1's.
- $\mathbf{G} = (\mathbf{C}\mathbf{X}_I)\mathbf{S}^{-1}(\mathbf{C}\mathbf{X}_I)^\top$ is the Mahalanobis Gram matrix
- $\mathbf{C} = \mathbf{I} - \frac{1}{I}\mathbf{1}\mathbf{1}^\top$ is the centering operator, $\mathbf{I}$ is an identity matrix of size I

association-based distance

Association-based for categorical: **total variation distance (TVD)**¹

The distance matrix $\mathbf{D}_{tvd}$ can be defined via the **delta framework**² upon properly defining the block-diagonal matrix $\mathbf{\Delta}$

Let $\mathbf{X}_{cat}$ be $I \times Q_c$ a data matrix of n observations described by Q_c categorical variables.

$$\mathbf{D} = \mathbf{Z}\mathbf{\Delta}\mathbf{Z}^T = [\mathbf{Z}_1 \quad \dots \quad \mathbf{Z}_{Q_c}] \begin{bmatrix} \mathbf{\Delta}_1 & & \\ & \ddots & \\ & & \mathbf{\Delta}_{Q_c} \end{bmatrix} \begin{bmatrix} \mathbf{Z}_1^T \\ \vdots \\ \mathbf{Z}_{Q_c}^T \end{bmatrix}$$

- in the framework, setting Δ_j determines the categorical distance measure of choice (independent- or association-based)

association-based distance

Association-based for categorical: **total variation distance (TVD)**¹ (2)

Consider the empirical **joint probability** distributions stored in the off-diagonal blocks of $\mathbf{P}$:

$$\mathbf{P} = \frac{1}{n} \begin{bmatrix} \mathbf{Z}_1^\top \mathbf{Z}_1 & \mathbf{Z}_1^\top \mathbf{Z}_2 & \cdots & \mathbf{Z}_1^\top \mathbf{Z}_{Q_c} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{Z}_{Q_c}^\top \mathbf{Z}_1 & \mathbf{Z}_{Q_c}^\top \mathbf{Z}_2 & \cdots & \mathbf{Z}_{Q_c}^\top \mathbf{Z}_{Q_c} \end{bmatrix}.$$

The block matrix $\mathbf{R}$ refer to the **conditional probability** distributions for each variable j given each variable i ($i, j = 1, \dots, Q_c, i \neq j$), stored in the block matrix

$$\mathbf{R} = \mathbf{P}_z^{-1}(\mathbf{P} - \mathbf{P}_z).$$

where $\mathbf{P}_z = \mathbf{P} \odot \mathbf{I}_{Q^*}$, and $\mathbf{I}_{Q^*}$ is the $Q^* \times Q^*$ identity matrix.

association-based distance

Association-based for categorical: **total variation distance (TVD)**¹ (3)

Let $\mathbf{r}_a^{ji}$ and $\mathbf{r}_b^{ji}$ be the rows of $\mathbf{R}_{ji}$, the (j, i) th off-diagonal block of $\mathbf{R}$.

The category dissimilarity between a and b for variable j based on the total variation distance (TVD) is defined as

$$\delta_{tvd}^j(a, b) = \sum_{i \neq j}^{Q_c} w_{ji} \Phi^{ji}(\mathbf{r}_a^{ji}, \mathbf{r}_b^{ji}) = \sum_{i \neq j}^{Q_c} w_{ji} \left[\frac{1}{2} \sum_{\ell=1}^{q_i} |\mathbf{r}_{a\ell}^{ji} - \mathbf{r}_{b\ell}^{ji}| \right],$$

where $w_{ji} = 1/(Q_c - 1)$ for equal weighting (can be user-defined).

TVD-based dissimilarity matrix is, therefore,

$$\mathbf{D}_{tvd} = \mathbf{Z} \Delta^{(tvd)} \mathbf{Z}^T.$$

association-based distance: a small example

Data

Consider two categorical variables:

- X_1 with categories A, B, C
- X_2 with categories u, v

```
# A tibble: 10 × 3
  id X1  X2
<int> <fct> <fct>
1     1  A    u
2     2  A    u
3     3  A    v
4     4  B    u
5     5  B    u
6     6  B    v
7     7  C    u
8     8  C    v
9     9  C    v
10    10  C    v
```

Indicator matrices

$$\mathbf{Z}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{Z}_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

association-based distance: from $\mathbf{Z}$ to $\mathbf{P}$

Let

$$\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2].$$

The empirical co-occurrence matrix is

$$\mathbf{P} = \frac{1}{10} \mathbf{Z}^\top \mathbf{Z}.$$

For this example,

$$\mathbf{P} = \begin{pmatrix} 0.30 & 0 & 0 & 0.20 & 0.10 \\ 0 & 0.30 & 0 & 0.20 & 0.10 \\ 0 & 0 & 0.40 & 0.10 & 0.30 \\ 0.20 & 0.20 & 0.10 & 0.50 & 0 \\ 0.10 & 0.10 & 0.30 & 0 & 0.50 \end{pmatrix}.$$

diagonal blocks contain marginal information; off-diagonal blocks contain joint proportions

association-based distance: from $\mathbf{P}$ to $\mathbf{R}$

The diagonal part of $\mathbf{P}$ is

$$\mathbf{P}_z = \mathbf{P} \odot \mathbf{I}_{Q^*} = \text{diag}(0.30, 0.30, 0.40, 0.50, 0.50).$$

The block matrix of conditional profiles is

$$\mathbf{R} = \mathbf{P}_z^{-1}(\mathbf{P} - \mathbf{P}_z).$$

For this example,

$$\mathbf{R} = \begin{pmatrix} 0 & 0 & 0 & 0.67 & 0.33 \\ 0 & 0 & 0 & 0.67 & 0.33 \\ 0 & 0 & 0 & 0.25 & 0.75 \\ 0.40 & 0.40 & 0.20 & 0 & 0 \\ 0.20 & 0.20 & 0.60 & 0 & 0 \end{pmatrix}.$$

each off-diagonal block contains conditional profiles across variables

association-based distance: from Δ to $\mathbf{D}$

We collect the category dissimilarity matrices in a block-diagonal matrix:

$$\Delta^{(tvd)} = \begin{pmatrix} \Delta_1^{(tvd)} & 0 \\ 0 & \Delta_2^{(tvd)} \end{pmatrix}.$$

The observation-level categorical distance matrix is then

$$\mathbf{D}_{tvd} = \mathbf{Z}\Delta^{(tvd)}\mathbf{Z}^\top = [\mathbf{Z}_1 \quad \mathbf{Z}_2] \begin{pmatrix} \Delta_1^{(tvd)} & 0 \\ 0 & \Delta_2^{(tvd)} \end{pmatrix} \begin{bmatrix} \mathbf{Z}_1^\top \\ \mathbf{Z}_2^\top \end{bmatrix}.$$

Equivalently,

$$\mathbf{D}_{tvd} = \mathbf{Z}_1\Delta_1^{(tvd)}\mathbf{Z}_1^\top + \mathbf{Z}_2\Delta_2^{(tvd)}\mathbf{Z}_2^\top.$$

category-level dissimilarities are translated into observation-level distances

Interaction-aware extensions

Cross-type interactions

Association-aware distances account for relations **within** variable blocks:

- continuous–continuous relations;
- categorical–categorical relations.

Cross-type structure

In mixed data, categorical differences may be meaningful because they are reflected in the continuous variables.

the next step is to make distances **interaction-aware**

How to measure interactions¹

Define Δ^{int} to account for continuous–categorical interactions and use it to augment Δ^{tvd} .

The mixed dissimilarity becomes

$$\mathbf{D}_{mix}^{(int)} = \mathbf{D}_{mah} + \mathbf{D}_{cat}^{(int)}.$$

where

$$\mathbf{D}_{cat}^{(int)} = \mathbf{Z}\tilde{\Delta}\mathbf{Z}^\top$$

and

$$\tilde{\Delta} = (1 - \alpha)\Delta^{tvd} + \alpha\Delta^{int}, \quad \alpha = \frac{1}{Q_c}.$$

What is Δ^{int} ?

The entry $\delta_{int}^j(a, b)$ measures how much the continuous variables help discriminate between observations choosing category a and those choosing category b for categorical variable j .

Category-pair classification problem

For each pair (a, b) :

- use the continuous variables as predictors;
- classify observations belonging to categories a and b ;
- use a nearest-neighbour rule in the continuous space.

Computing Δ_j^{int}

For each categorical variable j and each category pair (a, b) :

1. use $\mathbf{D}_{mah}$ to identify neighbours among observations belonging to a or b ;
2. consider a proportion of neighbours, say $\pi_{nn} = 0.1$;
3. classify observations using a prior-corrected decision rule;
4. compute balanced accuracy.

$$\text{BAcc}^j(a, b) = \frac{1}{2} \left[\text{TPR}_a^j(a, b) + \text{TPR}_b^j(a, b) \right].$$

Map predictive performance onto a non-negative separability scale:

$$\delta_{int}^j(a, b) = \max \{0, 2 \text{BAcc}^j(a, b) - 1\} \in [0, 1].$$

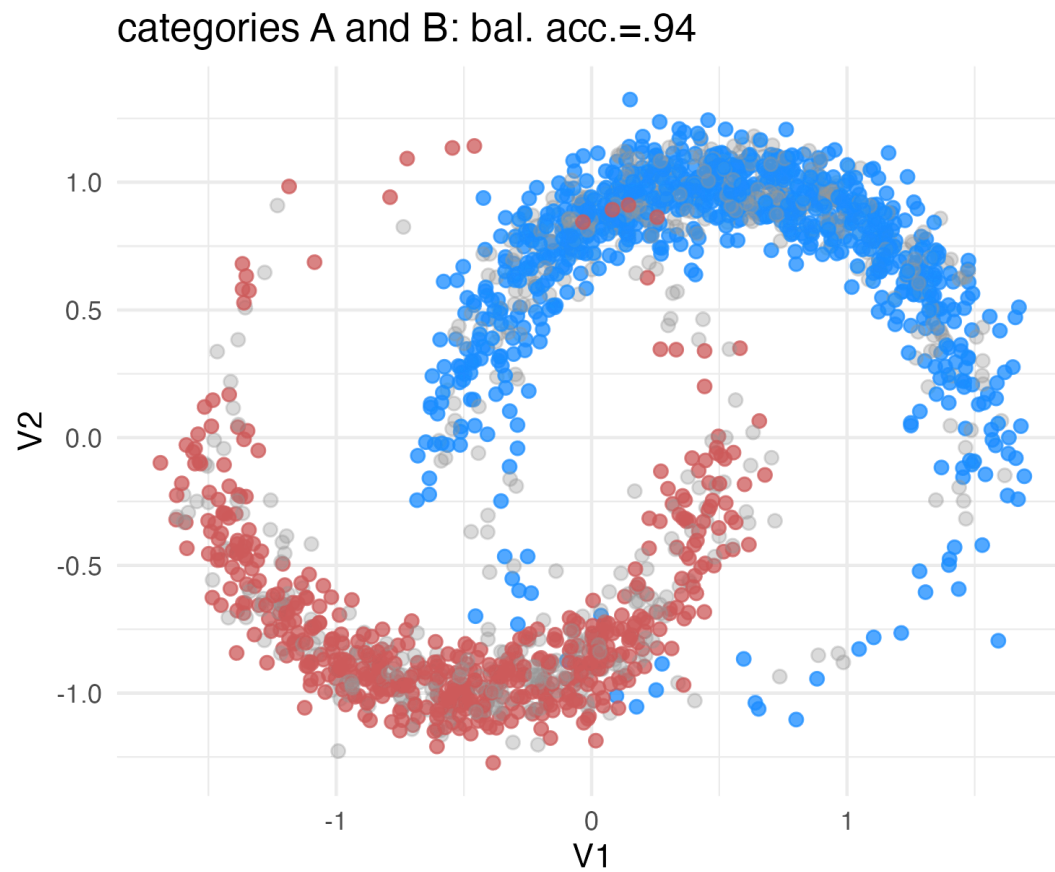
0 means chance-level or worse; 1 means perfect separation

Computing Δ_j^{int}

For categorical variable j with q_j categories, compute $\frac{q_j(q_j-1)}{2}$ category-pair quantities.

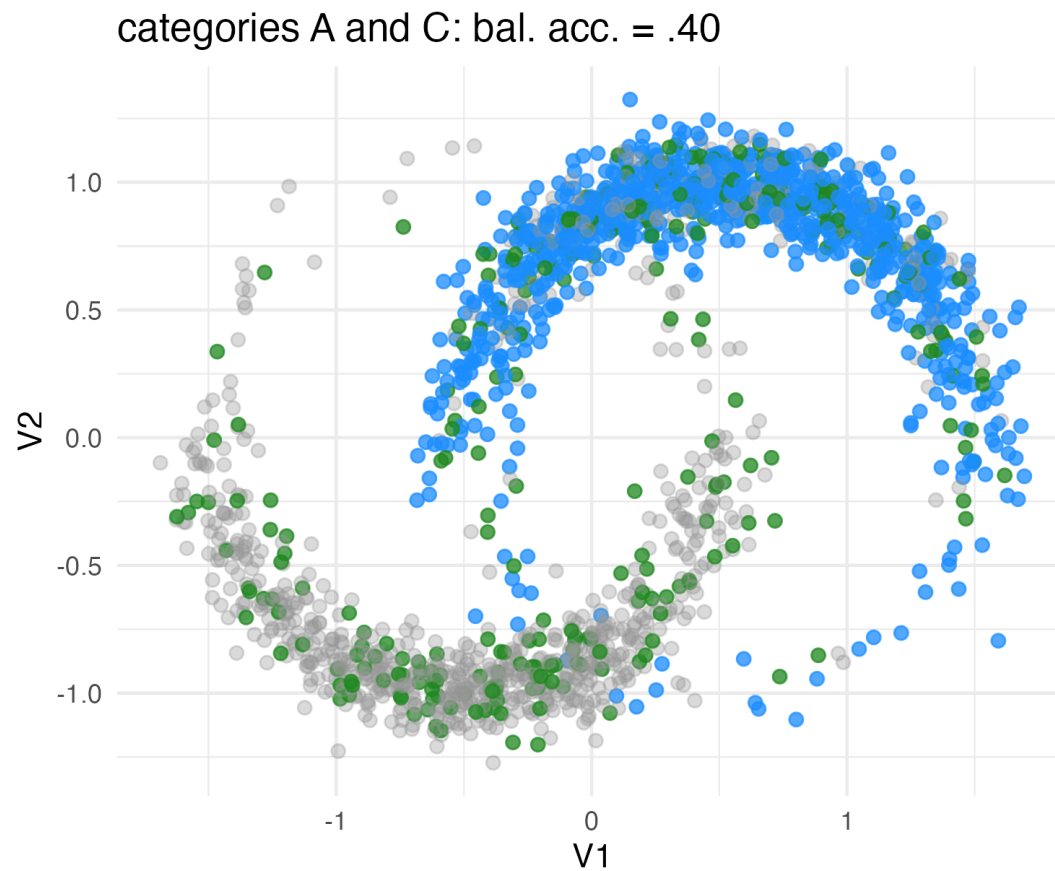
$$\Delta_{int} = \begin{pmatrix} 0 & \cdot & \cdot & \cdot \\ \cdot & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



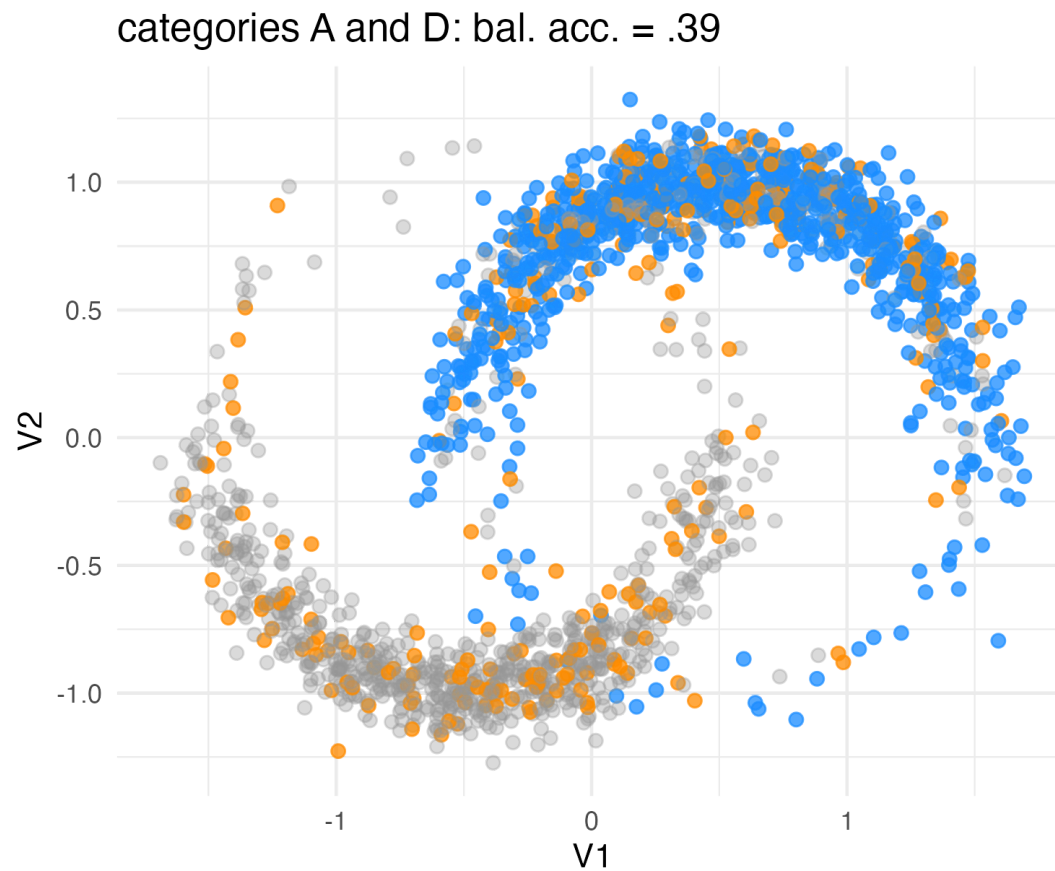
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \cdot & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



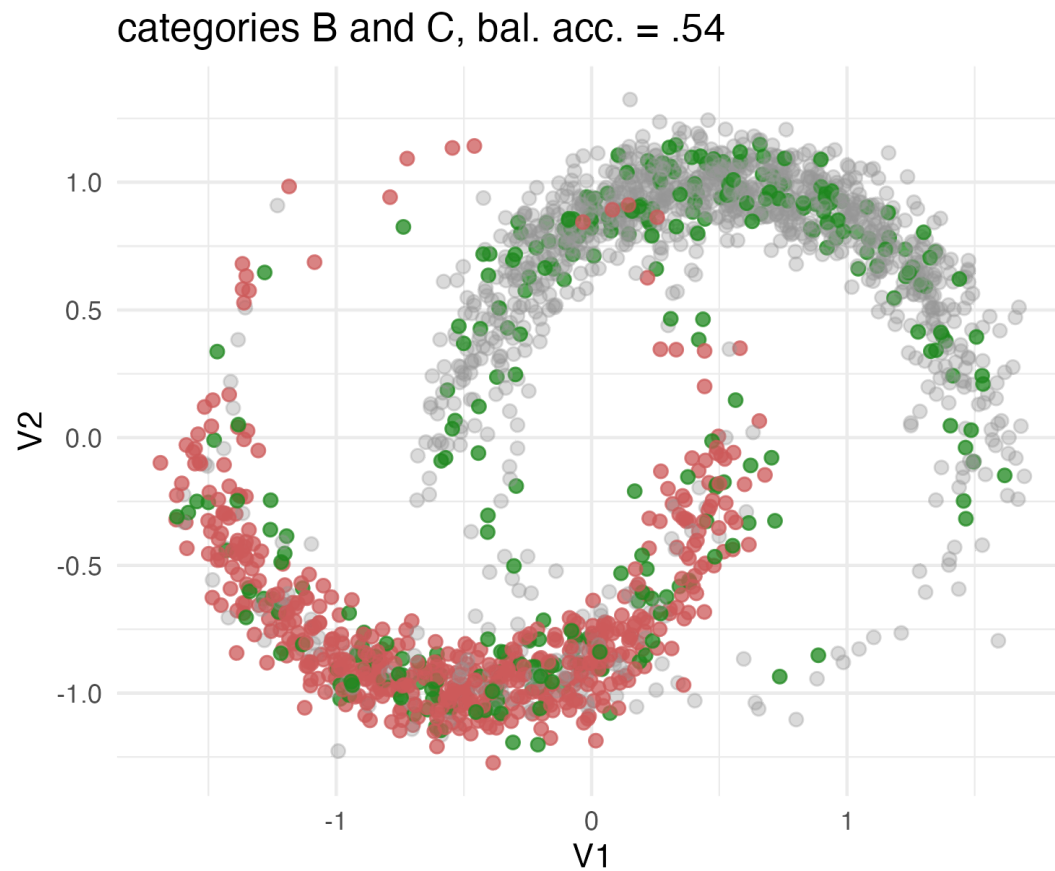
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \textcolor{red}{0.40} & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \textcolor{red}{0.40} & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



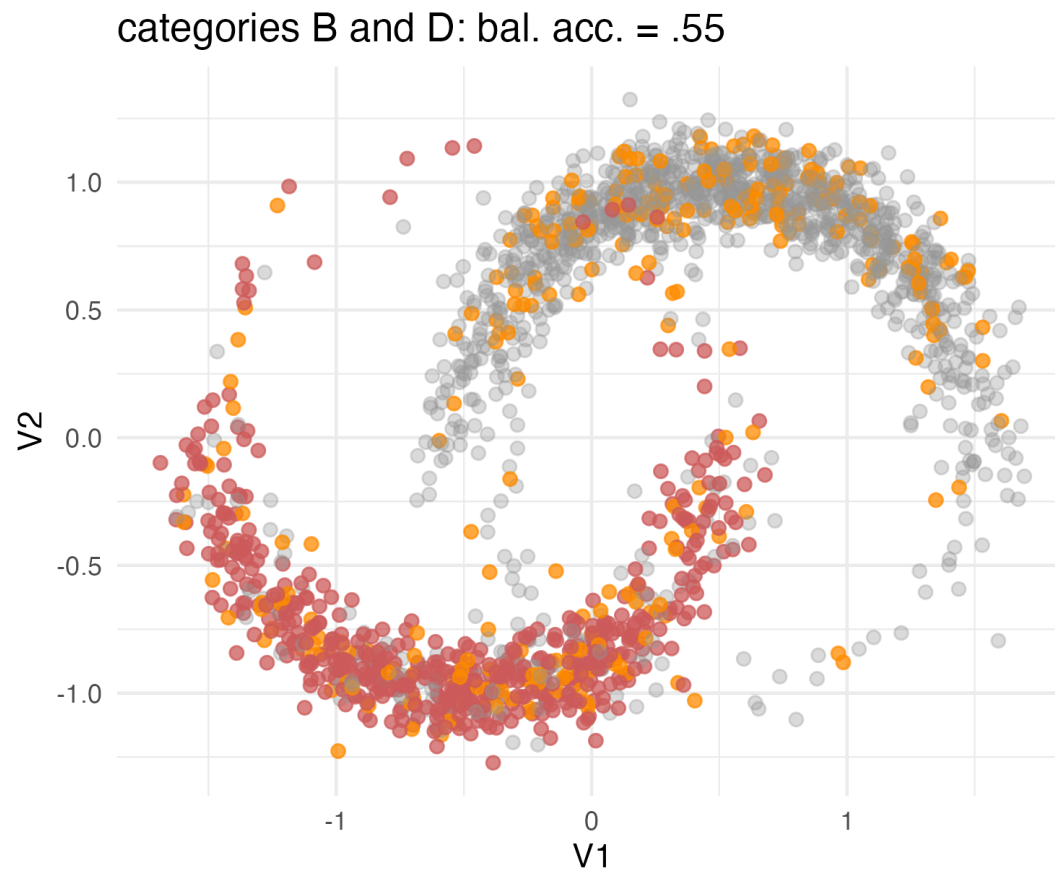
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & \cdot & \cdot \\ 0.40 & \cdot & 0 & \cdot \\ 0.39 & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



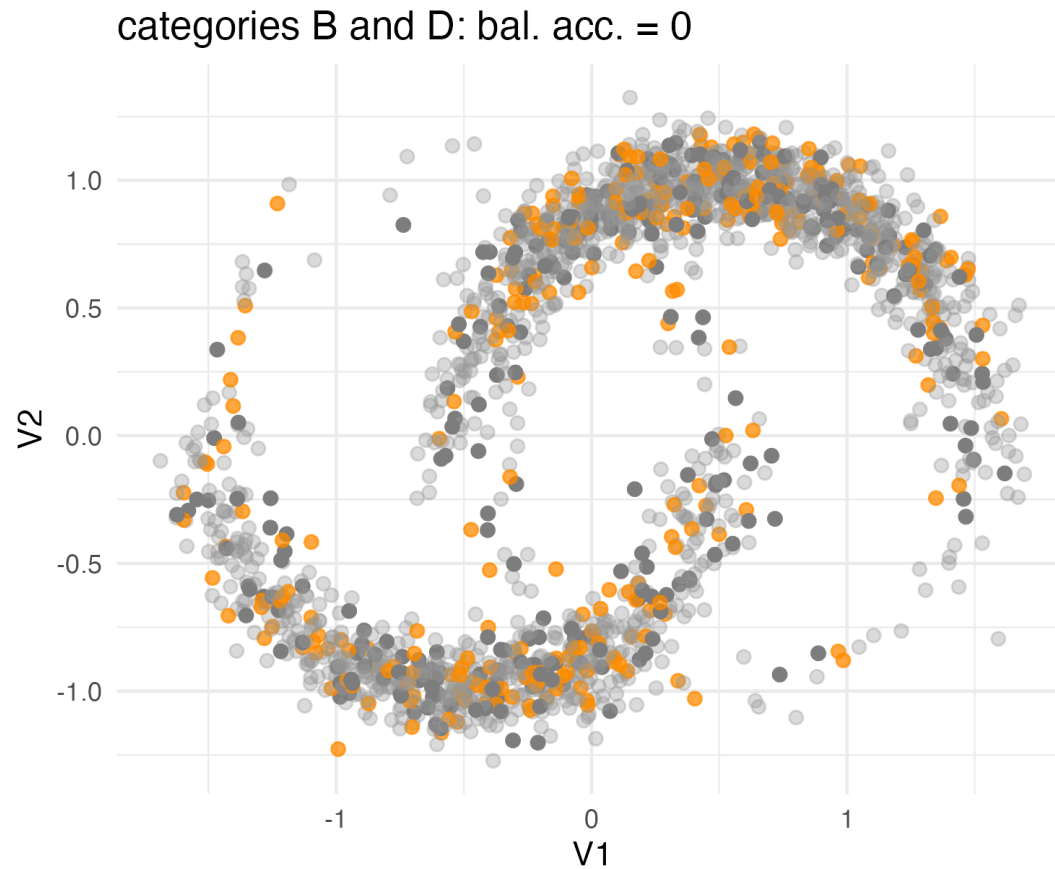
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & \cdot \\ 0.40 & 0.54 & 0 & \cdot \\ 0.39 & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & \textcolor{red}{0.55} \\ 0.40 & 0.54 & 0 & \cdot \\ 0.39 & \textcolor{red}{0.55} & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & 0.55 \\ 0.40 & 0.54 & 0 & 0 \\ 0.39 & 0.55 & 0 & 0 \end{pmatrix}$$

summarize category-pair separability in the continuous space

Just one-way interaction?

Factorization motivates the direction

Let $\mathbf{x}_i = (\mathbf{x}_{i_n}, \mathbf{x}_{i_c})$.

The joint distribution can always be written as

$$f(\mathbf{x}_{i_n}, \mathbf{x}_{i_c}) = f(\mathbf{x}_{i_n})f(\mathbf{x}_{i_c} \mid \mathbf{x}_{i_n}).$$

This motivates using the continuous variables to quantify categorical distinctions:

$$\mathbf{X}_n \longrightarrow \delta_{int}^j(a, b).$$

How can this result be connected to pairwise dissimilarities?

Just one-way interaction?

Dissimilarity and reference density

Starting from a non-negative dissimilarity from a prototype,

$$d_{\mathbf{S}_k}(\mathbf{x}_i, \mathbf{c}_k) \geq 0, \quad d_{\mathbf{S}_k}(\mathbf{c}_k, \mathbf{c}_k) = 0,$$

a distance-based probability model can be constructed, provided that the exponential kernel is normalisable ¹:

$$f(\mathbf{x}_i; \mathbf{c}_k, \mathbf{S}_k) = g(\mathbf{c}_k, \mathbf{S}_k, \eta) \exp \{-\eta d_{\mathbf{S}_k}(\mathbf{x}_i, \mathbf{c}_k)\}, \quad \eta > 0,$$

where $g(\mathbf{c}_k, \mathbf{S}_k, \eta)$ makes f a proper density.

Just one-way interaction?

Dissimilarity and reference density

Since the density is maximal at the reference point,

$$M_k = \max_{\mathbf{x}} f(\mathbf{x}; \mathbf{c}_k, \mathbf{S}_k) = f(\mathbf{c}_k; \mathbf{c}_k, \mathbf{S}_k) = g(\mathbf{c}_k, \mathbf{S}_k, \eta).$$

the **maximum-normalised kernel** is

$$\bar{K}_k(\mathbf{x}_i, \mathbf{c}_k) = \frac{f(\mathbf{x}_i; \mathbf{c}_k, \mathbf{S}_k)}{M_k} = \exp \{ -\eta d_{\mathbf{S}_k}(\mathbf{x}_i, \mathbf{c}_k) \}.$$

Therefore,

$$d_{\mathbf{S}_k}(\mathbf{x}_i, \mathbf{c}_k) = -\frac{1}{\eta} \log \bar{K}_k(\mathbf{x}_i, \mathbf{c}_k).$$

Just one-way interaction?

The pairwise continuous component

For pairwise comparison, let every observation act in turn as the reference prototype:

$$\mathbf{c}_k \longrightarrow \mathbf{x}_{i'_n}, \quad \mathbf{S}_k \longrightarrow \mathbf{S}.$$

Use the ordinary Mahalanobis distance and choose $\eta = 1$:

$$f_n(\mathbf{x}_{i_n}; \mathbf{x}_{i'_n}, \mathbf{S}) = g_n(\mathbf{S}) \exp \left\{ -d_{\text{mah}}(\mathbf{x}_{i_n}, \mathbf{x}_{i'_n}) \right\}.$$

Using the reference maximum $M_n = g_n(\mathbf{S})$ from the preceding construction
the **maximum-normalised continuous kernel** is

$$\overline{K}_n(i, i') = \frac{f_n(\mathbf{x}_{i_n}; \mathbf{x}_{i'_n}, \mathbf{S})}{M_n} = \exp \left\{ -d_{\text{mah}}(\mathbf{x}_{i_n}, \mathbf{x}_{i'_n}) \right\}.$$

$-\log \overline{K}_n(i, i')$ recovers the ordinary Mahalanobis distance

Just one-way interaction?

The pairwise interaction-aware categorical component

For categorical variable j , let $x_{ij_c} = a$ and $x_{i'j_c} = b$.

The interaction-aware category dissimilarity is

$$\tilde{\delta}^j(a, b) = (1 - \alpha)\delta_{tvd}^j(a, b) + \alpha\delta_{int}^j(a, b).$$

Assume that $\tilde{\delta}^j(a, b) \geq 0$ and $\tilde{\delta}^j(b, b) = 0$.

Choosing $\eta_j = 1$, define the categorical reference PMF

$$p_j(a; b) = \text{softmax}_{h \in \mathcal{A}_j} \left\{ -\tilde{\delta}^j(h, b) \right\}_a = \frac{\exp \left\{ -\tilde{\delta}^j(a, b) \right\}}{\sum_{h \in \mathcal{A}_j} \exp \left\{ -\tilde{\delta}^j(h, b) \right\}}.$$

smaller category dissimilarities receive larger reference probabilities

Just one-way interaction?

The pairwise interaction-aware categorical component

Because $\tilde{\delta}^j(b, b) = 0$, the maximum reference probability is

$$M_j(b) = p_j(b; b) = \frac{1}{\sum_{h \in \mathcal{A}_j} \exp \left\{ -\tilde{\delta}^j(h, b) \right\}}.$$

The **maximum-normalised categorical kernel**

$$\overline{K}_j(a, b) = \frac{p_j(a; b)}{M_j(b)} = \exp \left\{ -\tilde{\delta}^j(a, b) \right\},$$

$-\log \overline{K}_j(a, b) = \tilde{\delta}^j(a, b)$ recovers the interaction-aware dissimilarity

Just one-way interaction?

Combining the kernels

For the whole categorical vector, define the separable maximum-normalised kernel

$$\begin{aligned}\overline{K}_c^{(int)}(i, i') &= \prod_{j=1}^{Q_c} \overline{K}_j(x_{ij_c}, x_{i'j_c}) \\ &= \exp \left\{ - \sum_{j=1}^{Q_c} \tilde{\delta}^j(x_{ij_c}, x_{i'j_c}) \right\} \\ &= \exp \left\{ -d_c^{(int)}(\mathbf{x}_{i_c}, \mathbf{x}_{i'_c}) \right\}.\end{aligned}$$

Just one-way interaction?

Combining the kernels

Combine the continuous and categorical kernels:

$$\overline{K}_{\text{mix}}^{(int)}(i, i') = \overline{K}_{\text{n}}(i, i') \overline{K}_{\text{c}}^{(int)}(i, i') = \exp \left\{ -d_{\text{mah}}(\mathbf{x}_{i_n}, \mathbf{x}_{i'_n}) - d_{\text{c}}^{(int)}(\mathbf{x}_{i_c}, \mathbf{x}_{i'_c}) \right\}$$

$$-\log \overline{K}_{\text{mix}}^{(int)}(i, i') = d_{\text{mah}}(\mathbf{x}_{i_n}, \mathbf{x}_{i'_n}) + d_{\text{c}}^{(int)}(\mathbf{x}_{i_c}, \mathbf{x}_{i'_c})$$

recovers the overall dissimilarity

in algebraic form

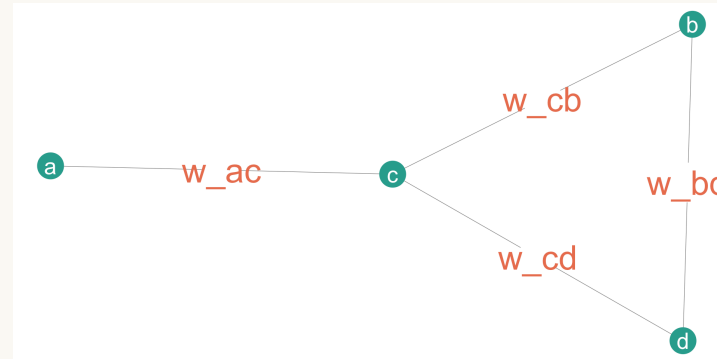
$$\mathbf{D}_{\text{mix}}^{(int)} = \mathbf{D}_{\text{mah}} + \mathbf{Z} \tilde{\Delta} \mathbf{Z}^{\top}.$$

downstream analysis

Spectral clustering: a graph partitioning problem

Graph representation

A graph representation of the data matrix $\mathbf{X}$: the aim is to cut it into K groups, or clusters.



The affinity matrix $\mathbf{A}$

The elements w_{ij} of $\mathbf{A}$ are high when observations i and j are likely to belong to the same group, and low otherwise.

Spectral clustering: making the graph easy to cut

An approximate solution to the graph partitioning problem (A. Ng, Jordan, & Weiss, 2001):

From distances to affinities

Start from the pairwise distance matrix $\mathbf{D}$ and build the affinity matrix

$$\mathbf{A} = \exp \left(-\frac{\mathbf{D}^2}{2\sigma^2} \right), \quad a_{ii} = 0.$$

The parameter σ controls the neighbourhood scale.

Normalized graph Laplacian

$$\mathbf{L} = \mathbf{D}_r^{-1/2} \mathbf{A} \mathbf{D}_r^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top,$$

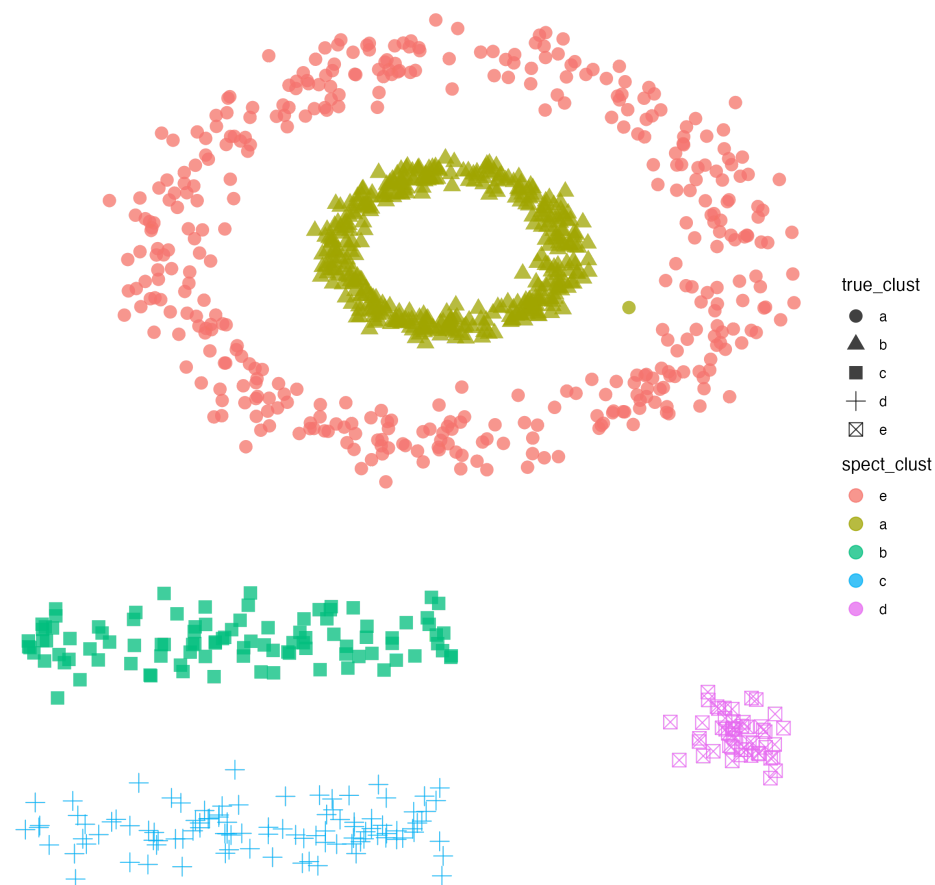
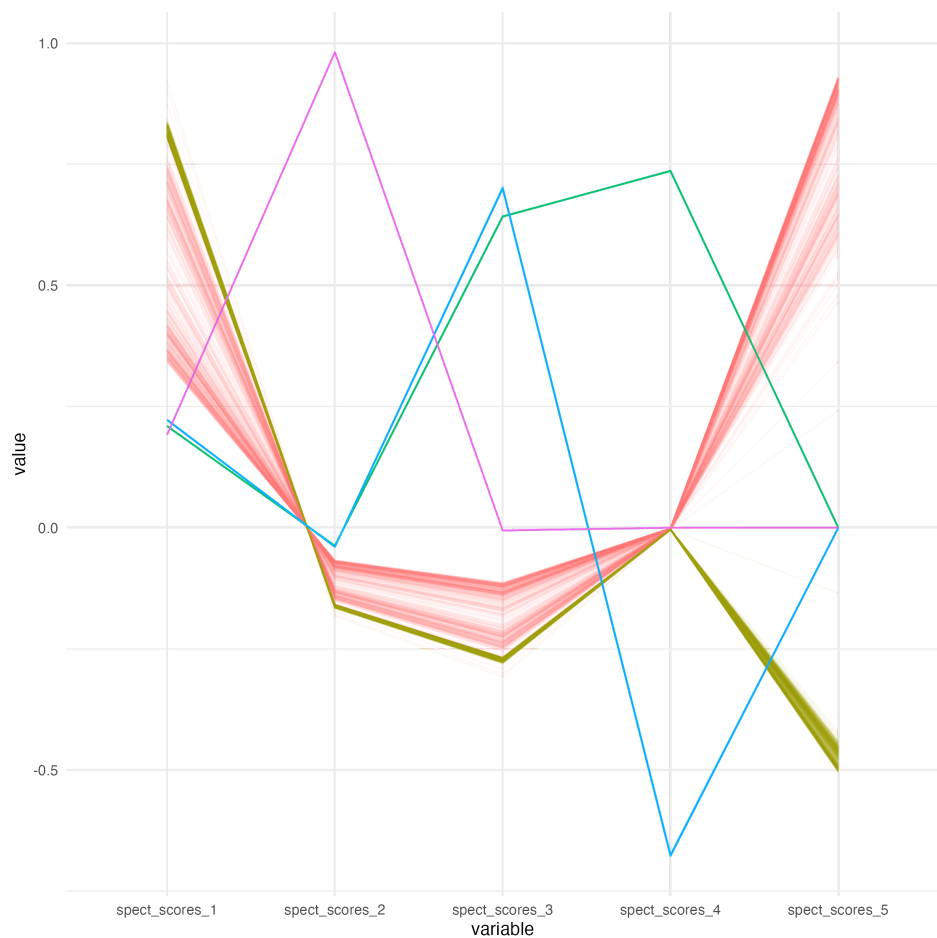
where $\mathbf{D}_r = \text{diag}(\mathbf{r})$, $\mathbf{r} = \mathbf{A} \mathbf{1}$, $\mathbf{1}$ is an I -dimensional vector of ones.

Spectral embedding

apply K -means to the rows of $\tilde{\mathbf{Q}}$, containing the first K eigenvectors of $\mathbf{L}$.

Why spectral clustering here?

Interaction-aware distances can encode local connectivity and non-convex structure.



Synthetic experiments

Simulation 1: associated mixed variables

General design

- two sample sizes: $I = 1000$ and $I = 2000$
- number of clusters fixed to $K = 3$
- 50 datasets generated for each configuration
- spectral clustering applied to each dissimilarity matrix
- performance measured by adjusted Rand index

Variable composition

Four scenarios were considered:

1. $Q_n = 15, Q_c = 15$
2. $Q_n = 10, Q_c = 20$
3. $Q_n = 20, Q_c = 10$
4. $Q_n = 10, Q_c = 10$ plus noise

Aim

The simulation creates mixed data with:

- correlated numerical variables;
- associated categorical variables;
- categorical variables depending on numerical variables;
- an additional noise scenario.

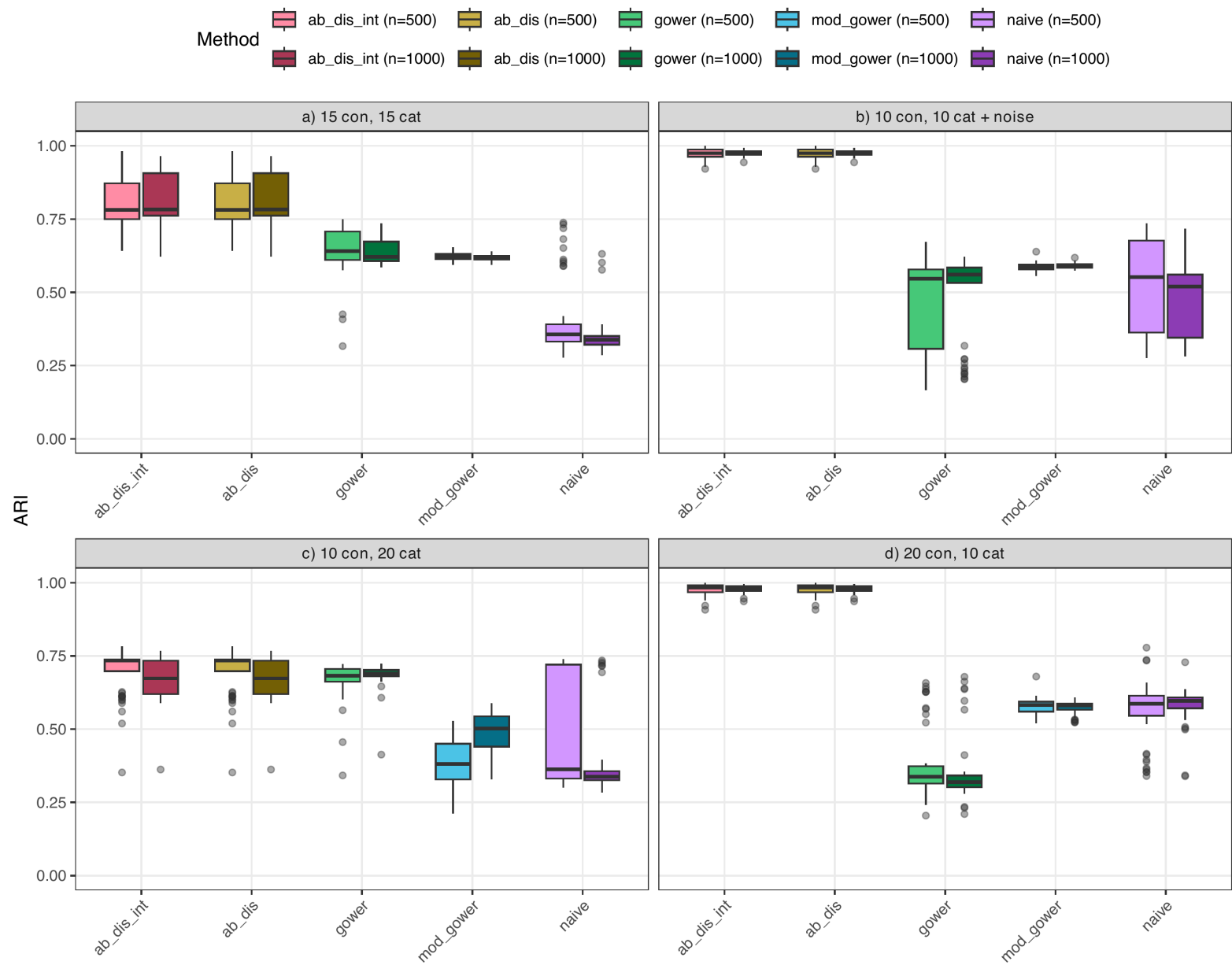
Simulation 1: associated mixed variables

Data-generating idea

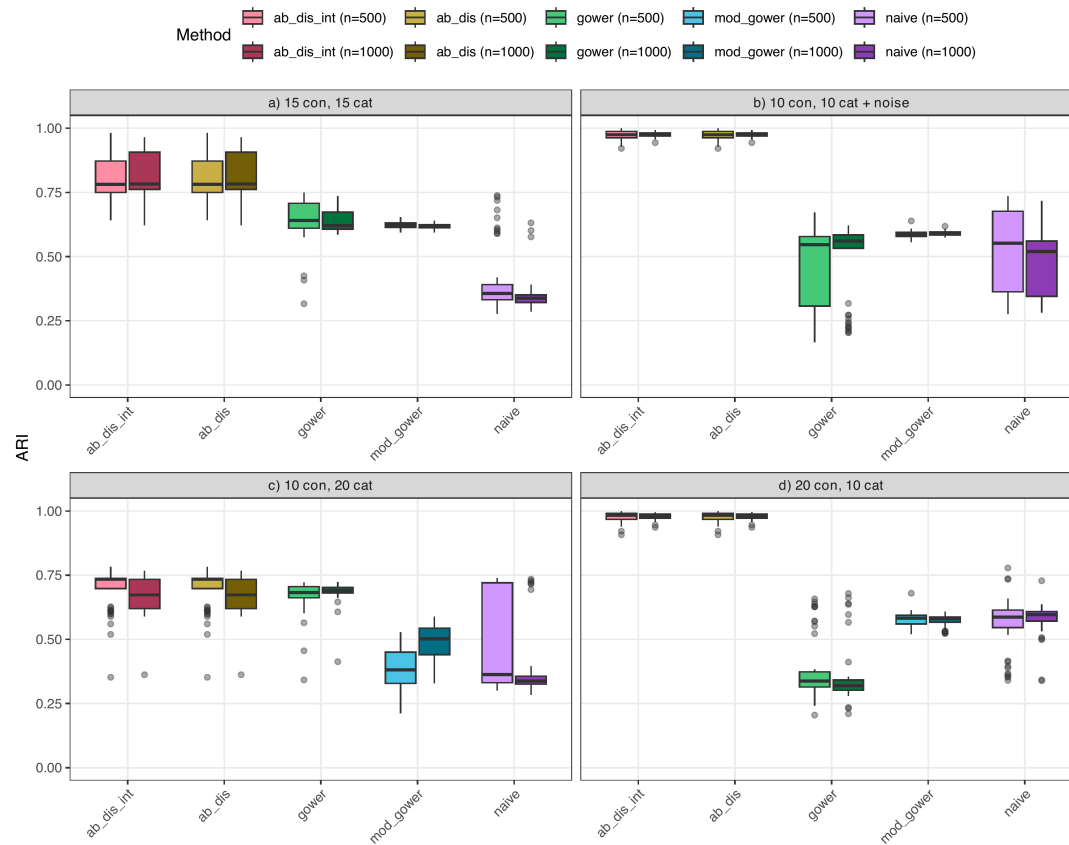
1. Generate numerical variables from cluster-specific multivariate normal distributions.
 - different cluster means;
 - different correlation structures.
2. Generate binary categorical variables conditionally on the numerical variables.
 - the first categorical variable depends on the numerical block;
 - later categorical variables depend on the numerical block and on previous categorical variables.

the design induces numerical correlations, categorical associations, and numerical–categorical dependence

Simulation 1: results



Simulation 1: results



- Association-based distances are generally the most accurate.
- When continuous variables dominate, AB distances achieve almost perfect recovery.
- Standard mixed-data distances are more sensitive to variable composition and noise.

Simulation 2: interaction-driven clusters

Design

- $I = 500, 1000$
- six continuous variables and three categorical variables
- V_4, V_5, V_6 generated independently from $N(0, 1)$
- V_1, V_2, V_3 generated conditionally on C_1 and C_2

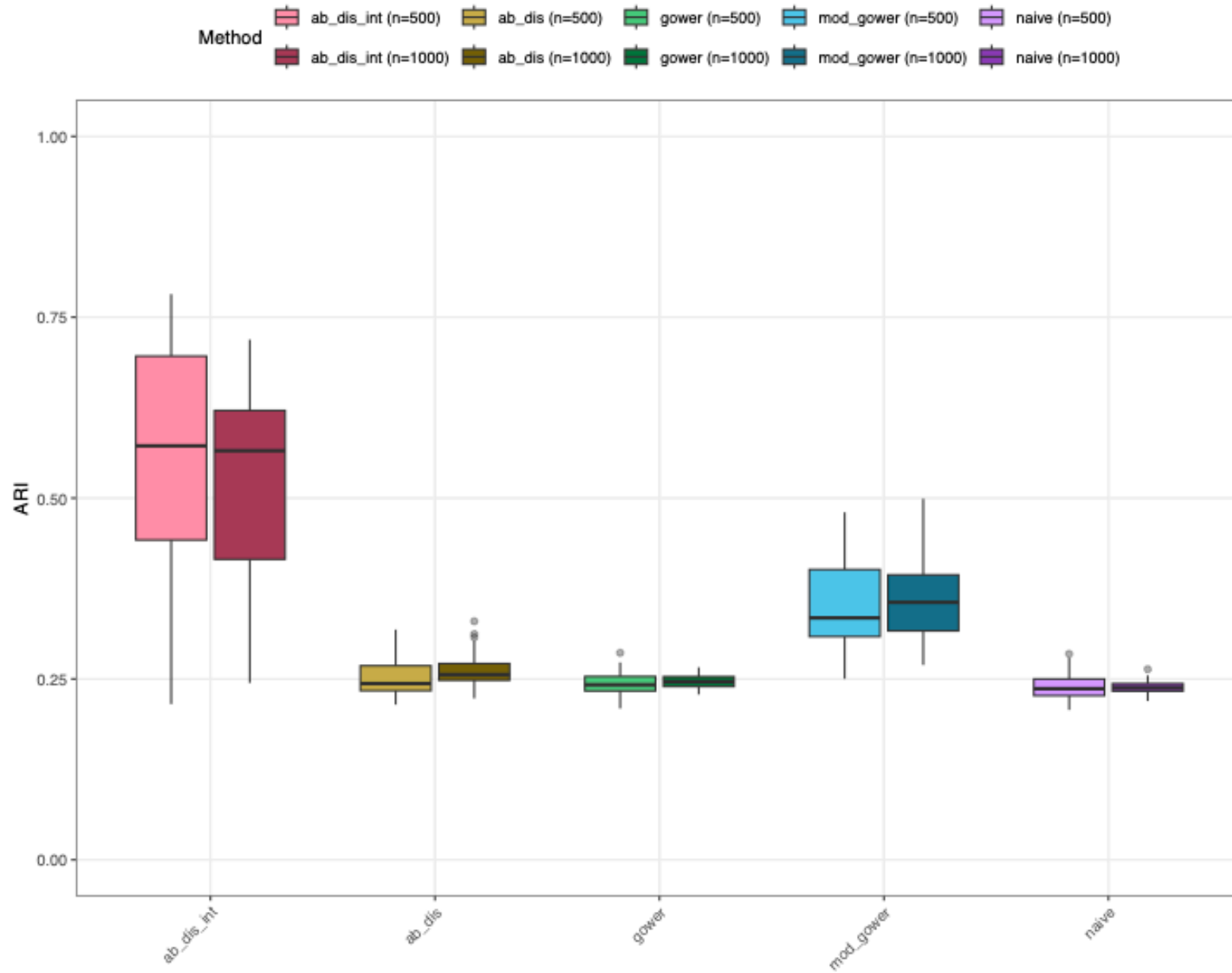
Main feature

The clusters are not defined by continuous variables alone or categorical variables alone.

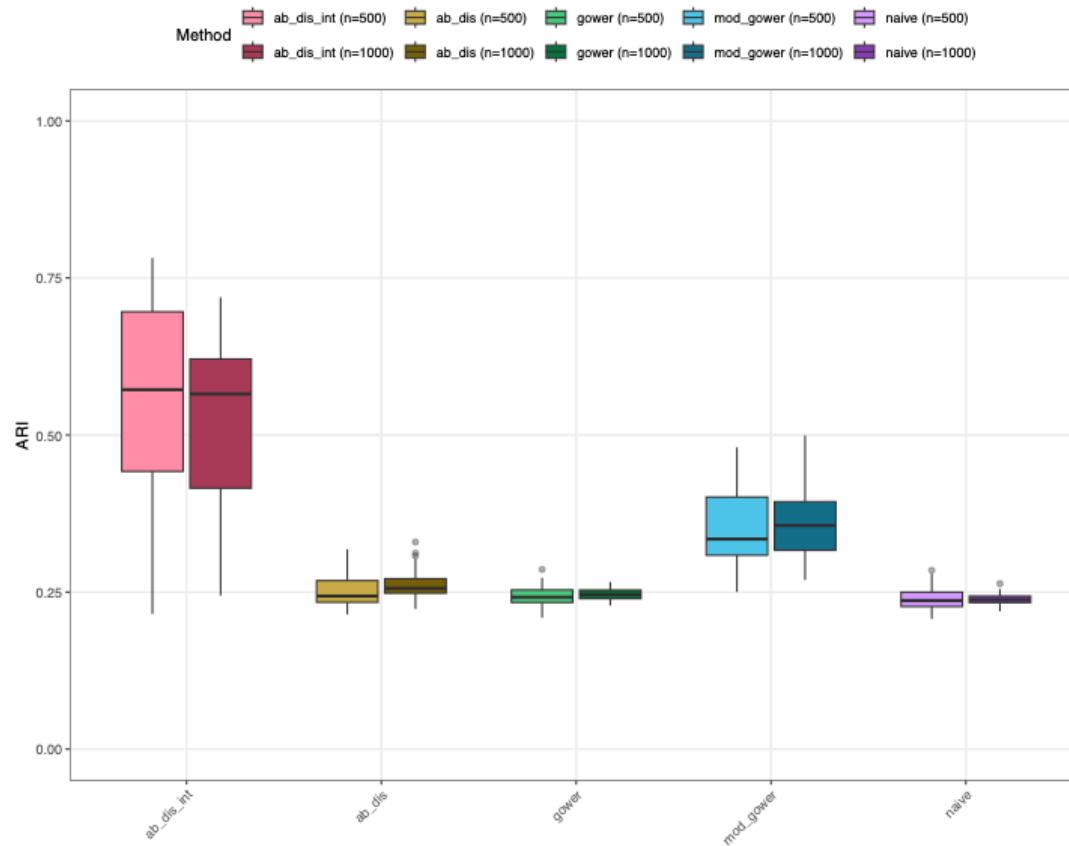
They are defined by their **cross-type interaction**.

this is the setting where Δ^{int} should matter

Simulation 2: results



Simulation 2: results



- `ab_dis_int` outperforms the competitors.
- distances that do not account for cross-type provide almost a chance-level performance.

Wrap-up

Association structure matters

Mahalanobis and TVD-based components account for redundancy within the numerical and categorical blocks.

Cross-type interactions must be encoded

Δ^{int} translates category separability in the numerical space into categorical dissimilarities; the maximum-normalised kernel representation makes the additive mixed construction coherent.

The distance determines what spectral clustering can recover

The simulations favour association-aware distances and show a clear gain from Δ^{int} when clusters are interaction-driven.

Main references

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Contact, and other stories

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manydist

Distance-Based Learning for Mixed-Type Data

`manydist` provides tools for constructing, computing, and using distance measures for numerical, categorical, and mixed-type data.



[Package website](#)

car^{me} 2027

22-25 june 2027 | Ionian University | Corfu, Greece

CARME 2027 is the tenth in a series of international conferences on multidimensional graphical techniques and the analysis of large data sets, especially categorical data. This conference will take place at the **Ionian University in Corfu, Greece**.



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Backup slides

Simulation 1: data generation

Numerical variables

Numerical variables are generated from cluster-specific multivariate normal distributions:

$$\mathbf{X}_n \mid C_k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

Cluster proportions:

$$P(C_1) = 0.20, \quad P(C_2) = 0.30, \quad P(C_3) = 0.50.$$

Cluster-specific structure:

- C_1 : means equal to 0, correlations 0.5;
- C_2 : means equal to 4, correlations 0.8;
- C_3 : means -1 for odd variables and 5 for even variables, correlations -0.5 .

All variances are set to 1.

Simulation 1: data generation

Categorical variables

Categorical variables are binary and generated recursively.

For categorical variable X_j^c ,

$$X_j^c \mid C_k \sim \text{Bernoulli}(p_{jk}),$$

with

$$p_{jk} = \left[1 + \exp \left(\boldsymbol{\beta}_k^\top \tilde{\mathbf{X}}_j \right) \right]^{-1}.$$

For $j = 1$,

$$\tilde{\mathbf{X}}_j = (1, \mathbf{X}_n),$$

while for $j > 1$,

$$\tilde{\mathbf{X}}_j = (1, \mathbf{X}_n, X_1^c, \dots, X_{j-1}^c).$$

The Gaussian special case

Gaussian reference model

Assume

$$\begin{aligned} f(\mathbf{x}_{i_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S}) \\ = \frac{1}{(2\pi)^{Q_d/2}} |\mathbf{S}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}}) \mathbf{S}^{-1} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}})^\top \right\}. \end{aligned}$$

The maximum occurs at

$$\mathbf{x}_{i_{con}} = \mathbf{x}_{i'_{con}},$$

and hence

$$M = f(\mathbf{x}_{i'_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S}) = \frac{1}{(2\pi)^{Q_d/2}} |\mathbf{S}|^{-1/2}.$$

The reference-normalized log-density is therefore

$$\begin{aligned} \log \frac{M}{f(\mathbf{x}_{i_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S})} &= \frac{1}{2} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}}) \mathbf{S}^{-1} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}})^{\top} \\ &= \boxed{\frac{1}{2} d_{\text{mah}}^2(\mathbf{x}_{i_{con}}, \mathbf{x}_{i'_{con}})}. \end{aligned}$$

Gaussian model: $\frac{1}{2} d_{\text{mah}}^2$

model used in the main derivation: d_{mah}